msvtx_trackSoftSumPt_NOSYS

MC vs data · overview · rank 65 / 98

rank 65 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    msvtx_trackSoftSumPtStats/sumPt
    MS1Vtx_sumTrackPt0p2Cone_raw
    MS1Vtx_barrel_sumTrackPt0p2Cone
    MS1Vtx_endcap_sumTrackPt0p2Cone

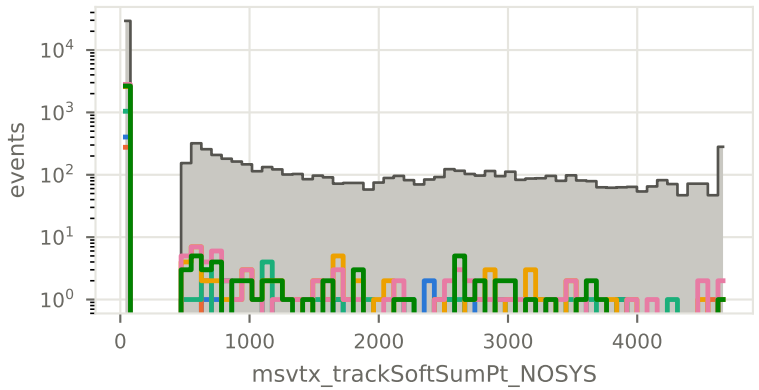
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.140	0.022	0.139	0.142	0.143	0.136	0.139	0.138	0.142	34,970	65/98
barrel	0.111	0.051	0.116	0.106	0.092	0.118	0.119	0.116	0.116	1,089	73/98
all	0.137	0.018	0.138	0.138	0.132	0.137	0.139	0.138	0.140	36,059	82/98

msvtx_trackSoftSumPt_NOSYS

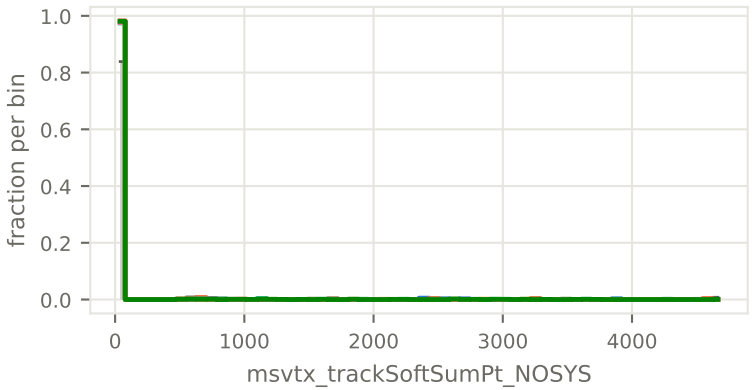
region: ENDCAP · rank 65 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.140 (noise 0.022) D_pooled 0.139 min 0.136 (mH200_mS50) max 0.143 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

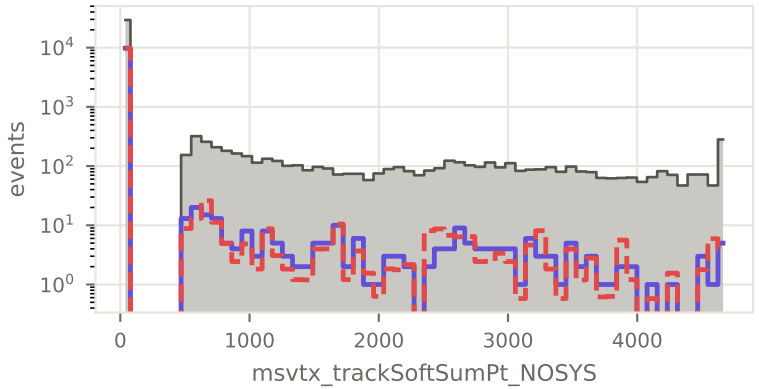


1b · each mass point vs background · normalized

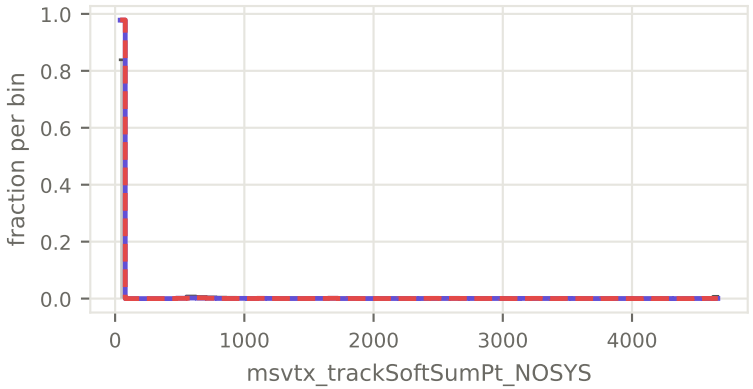


1 · each mass point
background n 34,970
mH125_mS15 D 0.142 n 411
mH125_mS55 D 0.143 n 281
mH200_mS50 D 0.136 n 1,071
mH400_mS100 D 0.139 n 2,648
mH600_mS150 D 0.138 n 2,882
mH1000_mS275 D 0.142 n 2,703

2a · combined signal vs background · raw

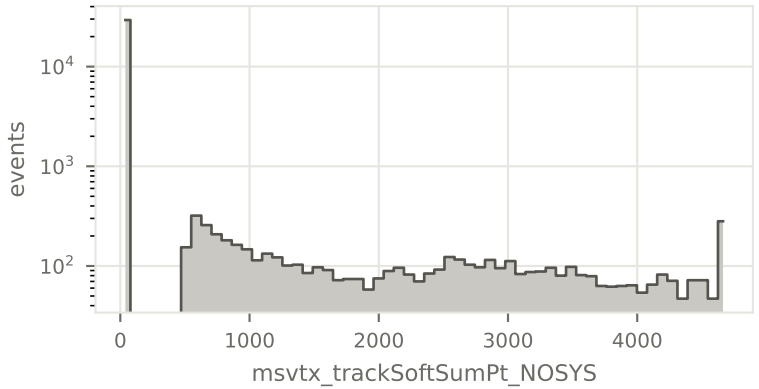


2b · combined signal vs background · normalized

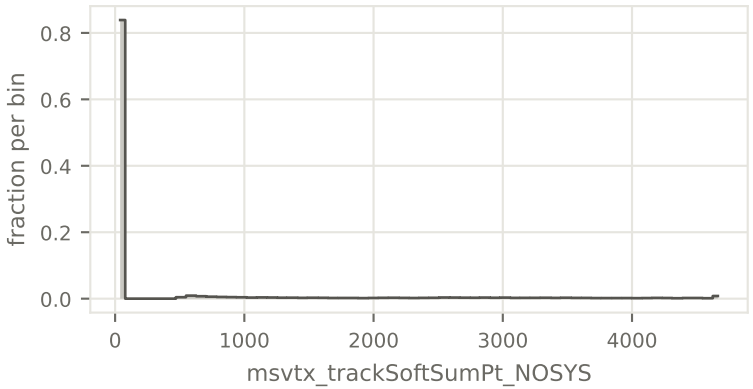


2 · combined signal
background n 34,970
combined, pooled D 0.139 n 9,996
equal weight per mass D 0.140

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

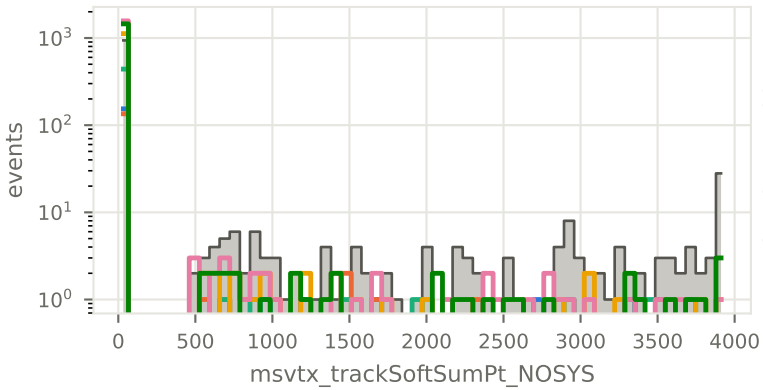
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_trackSoftSumPt_NOSYS

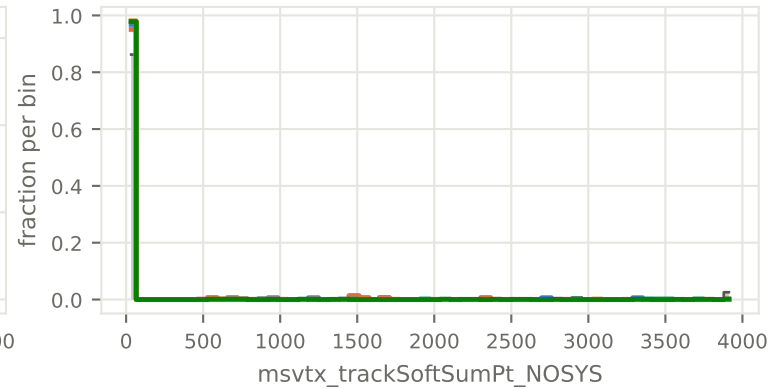
region: BARREL · rank 65 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.111 (noise 0.051) D_pooled 0.116 min 0.092 (mH125_mS55) max 0.119 (mH400_mS100) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

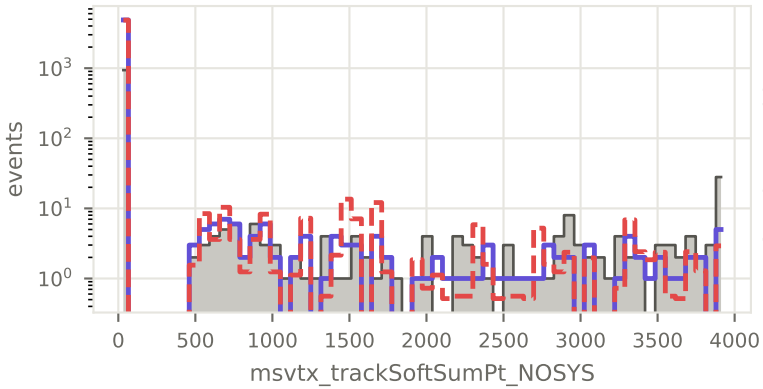


1b · each mass point vs background · normalized

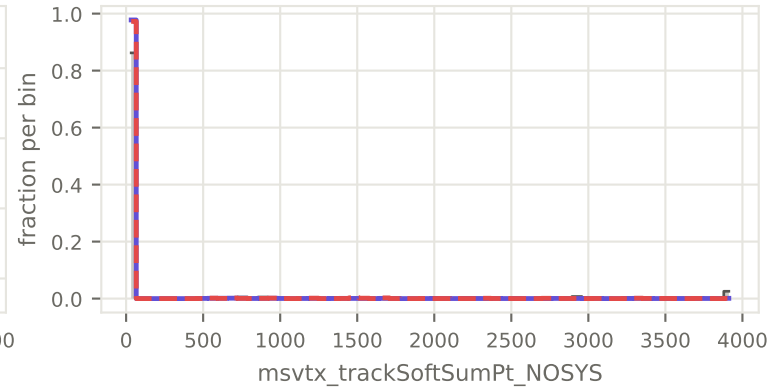


1 · each mass point
- background n 1,089
- mH125_mS15 D 0.106 n 159
- mH125_mS55 D 0.092 n 142
- mH200_mS50 D 0.118 n 449
- mH400_mS100 D 0.119 n 1,144
- mH600_mS150 D 0.116 n 1,608
- mH1000_mS275 D 0.116 n 1,485

2a · combined signal vs background · raw

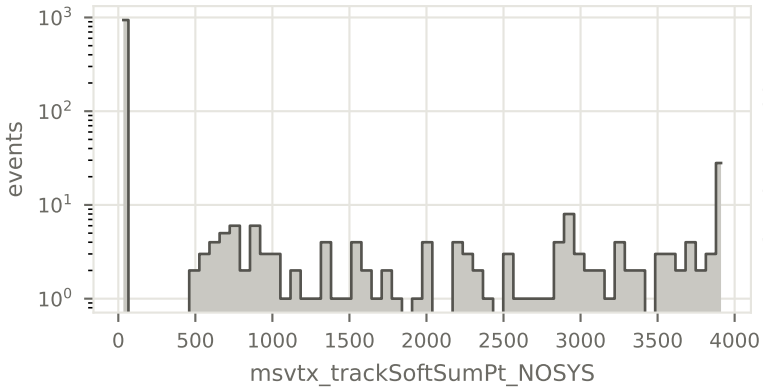


2b · combined signal vs background · normalized

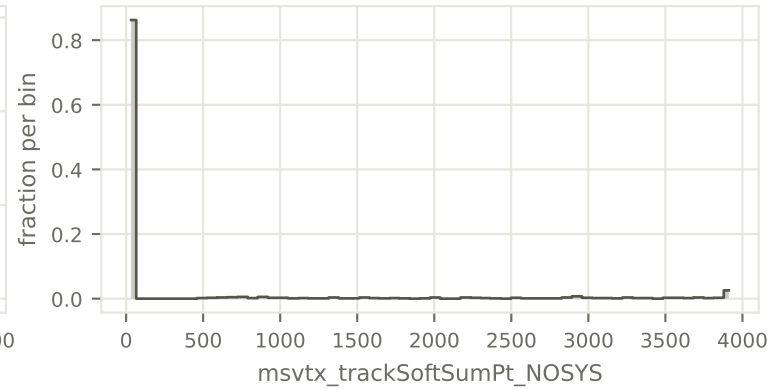


2 · combined signal
- background n 1,089
- combined, pooled D 0.116 n 4,987
- equal weight per mass D 0.111

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 1,089

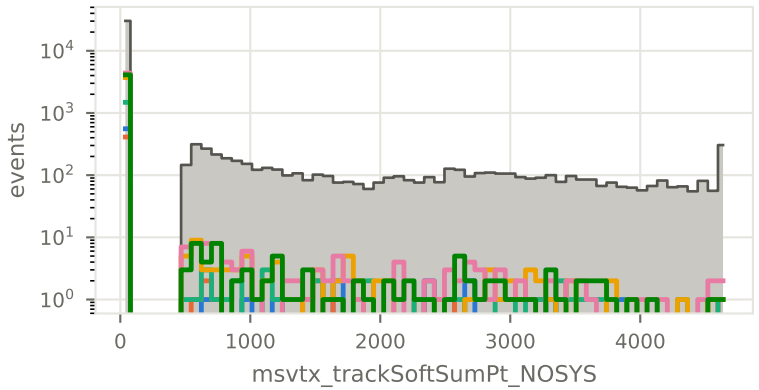
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_trackSoftSumPt_NOSYS

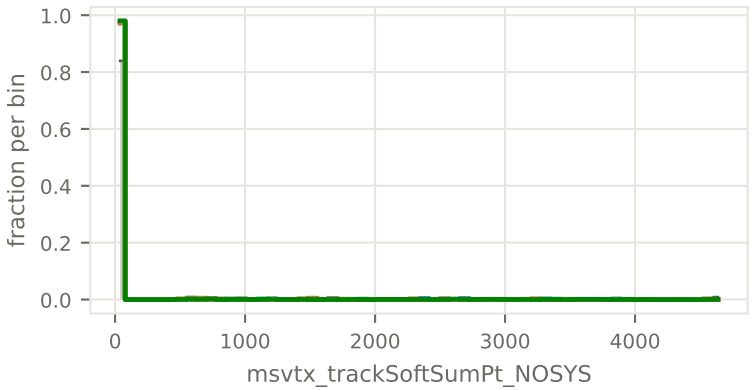
region: ALL · rank 65 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.137 (noise 0.018) D_pooled 0.138 min 0.132 (mH125_mS55) max 0.140 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

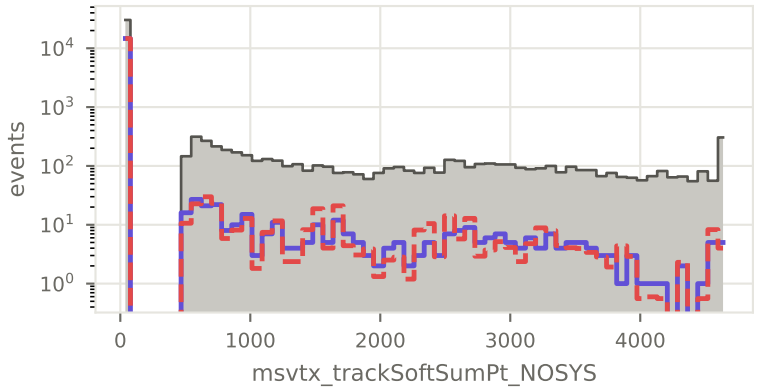


1b · each mass point vs background · normalized

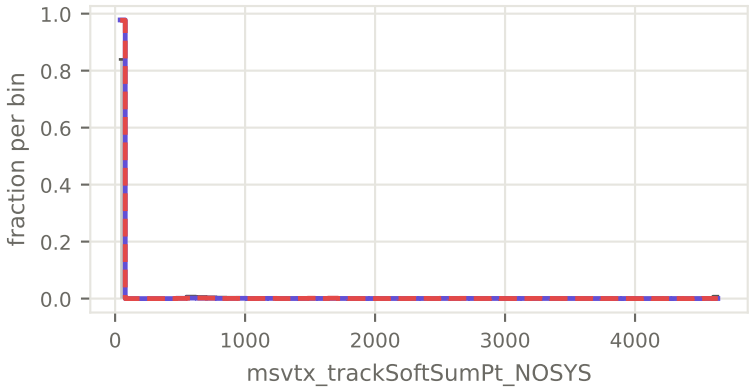


1 · each mass point
background n 36,059
mH125_mS15 D 0.138 n 570
mH125_mS55 D 0.132 n 423
mH200_mS50 D 0.137 n 1,520
mH400_mS100 D 0.139 n 3,792
mH600_mS150 D 0.138 n 4,490
mH1000_mS275 D 0.140 n 4,188

2a · combined signal vs background · raw

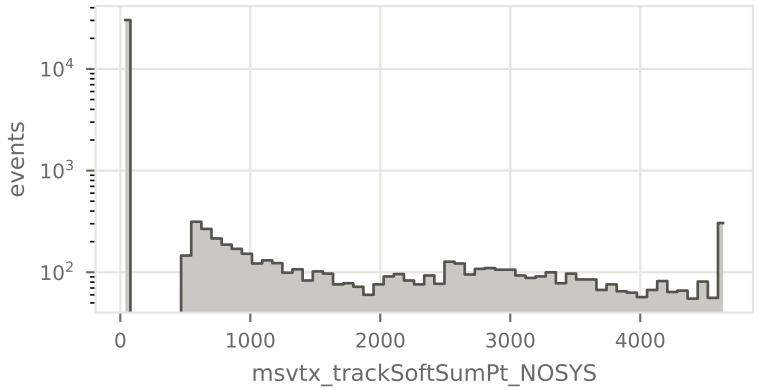


2b · combined signal vs background · normalized

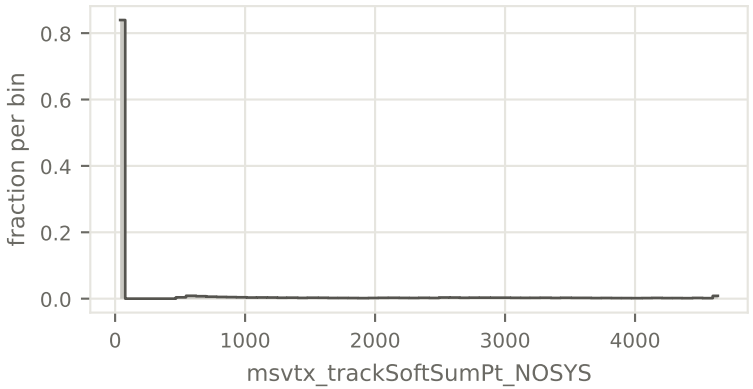


2 · combined signal
background n 36,059
combined, pooled D 0.138 n 14,983
equal weight per mass D 0.137

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed